

TreeLease

Rent a Mango Tree, Own the Harvest

Idea Contest Submission

Orbeetal Presents – Idea Contest

Submitted By

Shariyar Hossain Durjoy

Roll: 2003020

Department of Computer Science & Engineering (CSE)

Rajshahi University of Engineering & Technology (RUET)

Tagline

Connecting Urban Consumers with Local Orchards Through Technology



1. Executive Summary

TreeLease is a digital platform that allows people to rent a mango tree for an entire harvesting season directly from a registered orchard. Instead of purchasing mangoes from the market without knowing their origin or quality, customers can select a specific tree, reserve its seasonal harvest, and receive fresh mangoes through scheduled deliveries or by visiting the orchard during the harvesting period.

The platform creates value for both consumers and farmers. Consumers gain access to fresh, traceable, and naturally ripened mangoes while enjoying a unique experience of following the growth of "their" tree throughout the season. Farmers benefit from receiving advance payments before harvest, reducing their dependence on middlemen and uncertain market prices.

The idea is particularly feasible in **Rajshahi**, one of Bangladesh's largest mango-producing regions. As a student of **Rajshahi University of Engineering & Technology (RUET)**, I have observed the abundance of local mango orchards and the opportunity to connect them directly with customers through technology. This regional advantage makes the concept practical for initial implementation before expanding to other fruit-producing areas of Bangladesh.

TreeLease aims to transform the traditional fruit-selling process into a transparent, technology-driven agricultural service that benefits farmers, consumers, and the local economy. With a simple mobile application and partnerships with orchard owners, the platform has the potential to create a sustainable business model while promoting digital innovation in agriculture.

2. Problem Statement

Bangladesh is one of the major producers of mangoes, and Rajshahi is widely recognized for producing some of the country's finest varieties. Despite this, the journey from orchard to consumer remains inefficient and lacks transparency.

Most farmers rely on intermediaries to sell their produce, which often reduces their profit and creates uncertainty in pricing. At the same time, consumers usually have little information about where their mangoes come from, how they were grown, or whether they were naturally ripened.

Urban families who value fresh and authentic fruits also have no practical way to purchase directly from orchard owners or experience the harvesting process. This disconnect limits

trust between farmers and consumers while preventing farmers from building long-term relationships with their customers.

A digital platform that directly connects orchard owners with consumers can address these challenges by providing transparency, predictable income for farmers, and a unique farm-to-home experience for customers.

3. Proposed Solution

TreeLease is a digital platform that enables customers to reserve the seasonal harvest of a specific mango tree directly from a registered orchard.

At the beginning of the mango season, orchard owners list their available trees on the platform along with essential information such as tree location, mango variety, estimated harvest quantity, and rental price. Customers can browse these options, select a tree that matches their preferences, and complete the booking through the platform.

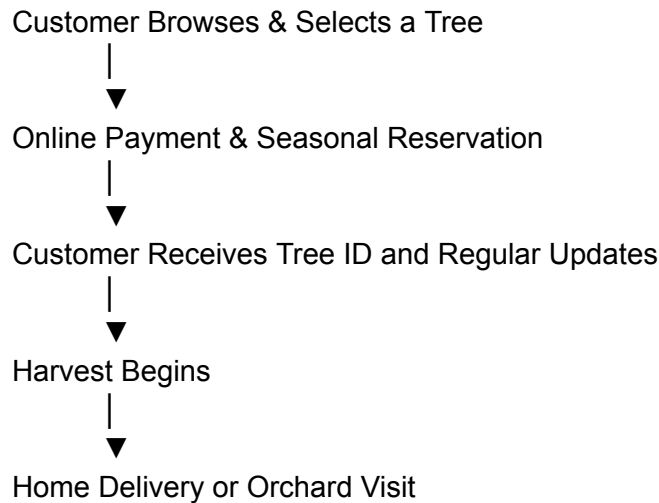
Once a tree is reserved, the customer receives a digital Tree ID and can follow the tree's progress through periodic photos and updates provided by the orchard owner. During the harvesting season, customers can either schedule home deliveries of freshly harvested mangoes or visit the orchard to collect the fruits themselves, depending on the services offered by the orchard.

By allowing customers to reserve an entire tree rather than purchasing mangoes from the market, TreeLease creates a direct relationship between consumers and farmers. Farmers receive payments in advance, improving financial stability and reducing dependence on intermediaries, while customers enjoy fresh, traceable, and authentic mangoes from a tree they have personally selected.

4. Key Features & Workflow

Workflow





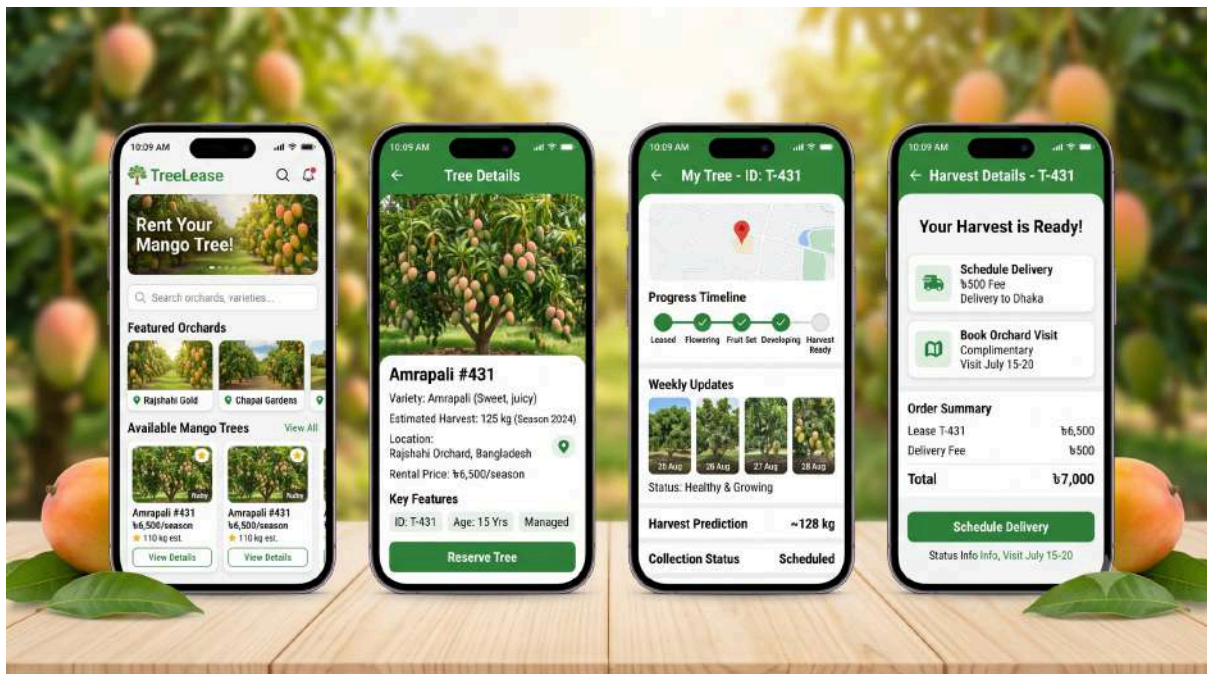
Key Features

- Browse available mango trees from registered orchards.
- View tree information, including mango variety, estimated harvest, and rental price.
- Reserve a tree for an entire harvesting season through online payment.
- Receive a unique digital Tree ID after successful booking.
- Track the tree's growth through periodic photos and updates.
- Choose between home delivery of harvested mangoes or collecting them directly from the orchard.
- Enable direct communication between customers and orchard owners, reducing dependence on intermediaries.



4. Key Features

- **Seasonal Tree Reservation:** Rent a specific mango tree for one harvesting season.
- **Tree Information:** View the tree's variety, estimated harvest, orchard location, and rental price before booking.
- **Tree Progress Updates:** Receive periodic photos and updates on the tree's growth throughout the season.
- **Flexible Harvest Options:** Choose home delivery of harvested mangoes or collect them directly from the orchard.
- **Direct Farmer Connection:** Buy directly from orchard owners, ensuring transparency and reducing dependence on intermediaries.
- **Digital Booking & Payment:** Reserve trees securely through an easy-to-use online platform.



5. Feasibility

Rajshahi is one of the largest mango-producing regions in Bangladesh, making it an ideal location to launch TreeLease. The abundance of local orchards, growing use of smartphones and digital payments, and the opportunity to connect farmers directly with consumers make the idea practical and achievable. Starting in Rajshahi allows the platform to validate the concept before expanding to other regions.

6. Business Model

TreeLease will generate revenue through seasonal tree rental fees, a small commission on each successful booking, optional delivery charges, and premium services such as orchard visits and priority customer support.

7. Future Scope

The platform can be expanded beyond mangoes to include other seasonal fruits such as litchi, guava, and jackfruit. In the future, features like live tree monitoring, AI-based harvest prediction, and international gifting can further enhance the user experience.

8. Conclusion

TreeLease is a simple yet innovative idea that creates a direct connection between farmers and consumers. By combining agriculture with digital technology, the platform offers farmers a more predictable source of income while providing customers with fresh, traceable mangoes and a unique orchard experience. Beginning in Rajshahi provides a strong foundation for developing and scaling this concept across Bangladesh.